Rogelio Rios

MIS 7375

October 12, 2026

Homework 3 Pseudocode

How I approached it

I built on the same patient registration form from Homework 1. The way I thought about it was pretty simple. Every field gets its own check function in script.js, and those functions run on the fly while the patient is typing or when they leave the field, so the warning shows up right away. Then the VALIDATE button calls every one of those same functions one more time and counts how many came back bad. If the count is zero, the Submit button shows up. If it is not zero, Submit stays hidden and the error messages stay on the page next to the fields that still need fixing. I left the HTML checks (required, minlength, maxlength, pattern) in the page as a backup, but the JavaScript is what does the real work now.

1. Helper functions (used by every check)

```
FUNCTION showError(fieldId, message)
    FIND the error span reserved under fieldId (id = fieldId + "_error")
    SET its text to message
    ADD the "invalid" class to the field so it gets the red border from style.css
END FUNCTION

FUNCTION clearError(fieldId)
```

FIND the error span under fieldId

SET its text to empty

REMOVE the "invalid" class from the field

END FUNCTION

Note: every field has an empty error span already sitting under it with a set min-height in style.css, so the form does not jump around when messages show up or go away.

2. VALIDATE button (main logic)

FUNCTION validateForm()

SET errorCount = 0

HIDE the Submit button

Your Information

IF checkFirstName() is false THEN errorCount = errorCount + 1

IF checkMiddleInit() is false THEN errorCount = errorCount + 1

IF checkLastName() is false THEN errorCount = errorCount + 1

IF checkDOB() is false THEN errorCount = errorCount + 1

IF checkSSN() is false THEN errorCount = errorCount + 1

IF checkLanguage() is false THEN errorCount = errorCount + 1

IF checkRequiredRadio("sex") is false THEN errorCount = errorCount + 1

Address and Contact

IF checkAddr1() is false THEN errorCount = errorCount + 1

IF checkAddr2() is false THEN errorCount = errorCount + 1

IF checkCity() is false THEN errorCount = errorCount + 1

IF checkState() is false THEN errorCount = errorCount + 1

IF checkZip() is false THEN errorCount = errorCount + 1

IF checkEmail() is false THEN errorCount = errorCount + 1

IF checkPhone() is false THEN errorCount = errorCount + 1

Insurance

IF checkRequiredRadio("has_insurance") is false THEN errorCount = errorCount + 1

IF checkInsuranceProvider() is false THEN errorCount = errorCount + 1

IF checkMemberId() is false THEN errorCount = errorCount + 1

Health History

IF checkHealthHistory() is false THEN errorCount = errorCount + 1

IF checkSymptoms() is false THEN errorCount = errorCount + 1

IF checkHealthScale() is false THEN errorCount = errorCount + 1

Patient Portal Account

IF checkUserId() is false THEN errorCount = errorCount + 1

IF checkPassword() is false THEN errorCount = errorCount + 1

IF checkPasswordConfirm() is false THEN errorCount = errorCount + 1

```
    Privacy Acknowledgment
    IF checkHipaa() is false THEN errorCount = errorCount + 1

    IF errorCount = 0 THEN
        DISPLAY "Everything looks good. You can submit now."
        SHOW the Submit button
    ELSE
        DISPLAY "Please fix the " + errorCount + " field(s) marked in red."
        KEEP the Submit button hidden
        MOVE focus to the first field that has an error
    END IF
END FUNCTION
```

Note: the checks do not stop at the first error. I want every bad field flagged in one pass so the patient is not fixing things one at a time.

3. Field checks

```
Your Information
FUNCTION checkFirstName()
    GET firstname, TRIM spaces
    IF value is empty THEN showError("First name is required."), RETURN false
    IF length > 30 THEN showError("First name must be 30 characters or less."), RETURN false
```

IF value has anything other than letters, apostrophes or dashes THEN

showError("Letters, apostrophes and dashes only."), RETURN false

END IF

clearError(), RETURN true

END FUNCTION

FUNCTION checkMiddleInit()

GET middleinit, TRIM spaces, CONVERT to uppercase, PUT it back in the field

IF value is empty THEN clearError(), RETURN true (optional)

IF value is not a single letter THEN showError("One letter only, no numbers."), RETURN false

clearError(), RETURN true

END FUNCTION

FUNCTION checkLastName()

Same logic as checkFirstName on lastname, with "Last name" in the messages.

END FUNCTION

FUNCTION formatDOB() (runs oninput)

REMOVE everything that is not a digit, CUT to 8 digits

INSERT a slash after the 2nd digit and after the 4th digit

PUT the formatted value back in the field

END FUNCTION

FUNCTION checkDOB()

GET dob, SPLIT on "/" into month, day, year

IF any piece is missing or not a number THEN showError("Use MM/DD/YYYY."),

RETURN false

IF month < 1 OR month > 12 THEN showError("Month must be 01 to 12."), RETURN false

BUILD a date from month, day, year

IF the built date's month or day does not match what was typed THEN

showError("That date does not exist."), RETURN false (catches things like 02/30 or 04/31)

END IF

IF date is after today THEN showError("Date of birth cannot be in the future."), RETURN false

IF date is more than 120 years before today THEN showError("Date of birth cannot be more than 120 years ago."), RETURN false

clearError(), RETURN true

END FUNCTION

FUNCTION formatSSN() (runs oninput)

REMOVE everything that is not a digit, CUT to 9 digits

IF more than 3 digits THEN insert a dash after the 3rd digit

IF more than 5 digits THEN insert a dash after the 5th digit

PUT the formatted value back in ssn_plain (type password, so it still shows dots)

END FUNCTION

FUNCTION checkSSN()

GET ssn_plain, REMOVE the dashes

IF value is empty THEN showError("SSN is required."), RETURN false

IF value is not exactly 9 digits THEN showError("SSN must be 9 numbers."), RETURN false

clearError(), RETURN true

END FUNCTION

FUNCTION checkLanguage()

GET preferred_language, TRIM spaces

IF value is empty THEN clearError(), RETURN true (optional)

IF length > 20 OR value has anything other than letters and spaces THEN

showError("Letters only, 20 characters max."), RETURN false

END IF

clearError(), RETURN true

END FUNCTION

FUNCTION checkRequiredRadio(groupName) (used for sex and has_insurance)

IF no option in groupName is checked THEN showError(groupName, "Please choose one."), RETURN false

clearError(groupName), RETURN true

END FUNCTION

Address and Contact

FUNCTION checkAddr1()

GET addr1, TRIM spaces

IF value is empty THEN showError("Address is required."), RETURN false

IF length < 2 OR length > 30 THEN showError("Address must be 2 to 30 characters."),

RETURN false

IF value has characters other than letters, numbers, spaces, periods, commas, dashes or #

THEN

showError("Please remove special characters."), RETURN false

END IF

clearError(), RETURN true

END FUNCTION

FUNCTION checkAddr2()

GET addr2, TRIM spaces

IF value is empty THEN clearError(), RETURN true (optional)

OTHERWISE run the same length and character checks as checkAddr1

END FUNCTION

FUNCTION checkCity()

GET city, TRIM spaces

```
    IF value is empty THEN showError("City is required."), RETURN false
    IF length < 2 OR length > 30 THEN showError("City must be 2 to 30 characters."),
    RETURN false
    IF value has anything other than letters, spaces, periods, apostrophes or dashes THEN
        showError("Letters only."), RETURN false
    END IF
    clearError(), RETURN true
END FUNCTION

FUNCTION checkState()
    IF the selected option is the blank "-- Select state --" option THEN
        showError("Please pick a state."), RETURN false
    END IF
    clearError(), RETURN true
END FUNCTION

FUNCTION checkZip()
    GET zip, TRIM spaces
    IF value is empty THEN showError("ZIP code is required."), RETURN false
    IF value is not exactly 5 digits THEN showError("ZIP code must be 5 numbers."), RETURN
    false
    clearError(), RETURN true
END FUNCTION
```

```
FUNCTION checkEmail()
    GET email, TRIM spaces, CONVERT to lowercase, PUT it back in the field
    IF value is empty THEN showError("Email is required."), RETURN false
    IF value does not match name@domain.tld THEN showError("Email must look like
    name@domain.com."), RETURN false
    clearError(), RETURN true
END FUNCTION

FUNCTION checkPhone()
    GET phone1, REMOVE everything that is not a digit
    GET the checked contact_pref value
    IF value is empty THEN
        IF contact_pref is Phone call (P) OR Text (T) THEN
            showError("We need a phone number to call or text you."), RETURN false
        END IF
        clearError(), RETURN true (otherwise optional)
    END IF
    IF value is not exactly 10 digits THEN showError("Phone must be 10 digits."), RETURN
    false
    IF the area code or the next 3 digits start with 0 or 1 THEN showError("That is not a valid
    US number."), RETURN false
    REFORMAT the field as (713) 555-0123
```

```
    clearError(), RETURN true
END FUNCTION


Insurance
FUNCTION checkInsuranceProvider()
    GET the checked has_insurance value and the insurance_provider selection
    IF has_insurance is Yes (Y) AND provider is blank THEN
        showError("Please pick your insurance provider."), RETURN false
    END IF
    clearError(), RETURN true
END FUNCTION


FUNCTION checkMemberId()
    GET member_id, TRIM spaces, CONVERT to uppercase
    IF value is empty THEN clearError(), RETURN true (optional)
    IF length > 20 OR value has anything other than letters, numbers or dashes THEN
        showError("Letters, numbers and dashes only, as shown on your card."), RETURN false
    END IF
    clearError(), RETURN true
END FUNCTION


FUNCTION toggleInsurance() (runs onchange on has_insurance)
    IF has_insurance is No (N) THEN
```

CLEAR and DISABLE insurance_provider and member_id, clearError on both

ELSE

ENABLE insurance_provider and member_id

END IF

END FUNCTION

Health History

FUNCTION checkHealthHistory()

IF hx_none is checked AND any other hx_ box is also checked THEN

showError("You checked None of the above along with a condition. Please pick one or the other."), RETURN false

END IF

clearError(), RETURN true

END FUNCTION

Note: zero boxes checked is a valid answer since it is "check all that apply." The flu_vaccinated, tobacco_use and contact_pref radio groups are optional, so nothing to check there beyond what the browser already does.

FUNCTION checkSymptoms()

GET symptoms

IF value is empty THEN clearError(), RETURN true (optional)

IF value contains < or > THEN showError("Please do not use < or >."), RETURN false

IF length > 500 THEN showError("Please keep it under 500 characters."), RETURN false

clearError(), RETURN true

END FUNCTION

FUNCTION updateHealthScale() (runs oninput, moved out of the HTML into script.js)

GET health_scale value

SET health_scale_output to that value so the patient sees the number as they slide

END FUNCTION

FUNCTION checkHealthScale()

IF value < 1 OR value > 10 THEN showError("Please pick a number from 1 to 10."),

RETURN false

clearError(), RETURN true

END FUNCTION

Patient Portal Account

FUNCTION checkUserId()

GET userid, TRIM spaces

IF value is empty THEN showError("User ID is required."), RETURN false

IF first character is a number THEN showError("User ID cannot start with a number."),

RETURN false

IF length < 5 OR length > 20 THEN showError("User ID must be 5 to 20 characters."),

RETURN false

```
    IF value has anything other than letters, numbers, dashes or underscores THEN
        showError("No spaces or special characters. Dash and underscore are ok."), RETURN
        false
    END IF
    clearError(), RETURN true
END FUNCTION

FUNCTION checkPassword()
    GET password_plain and userid
    SET missing = empty list
    IF length < 8 THEN ADD "at least 8 characters" to missing
    IF no uppercase letter THEN ADD "1 uppercase letter" to missing
    IF no lowercase letter THEN ADD "1 lowercase letter" to missing
    IF no number THEN ADD "1 number" to missing
    IF password equals userid (ignoring case) THEN ADD "cannot match your User ID" to
    missing
    IF missing is not empty THEN showError("Password needs: " + missing), RETURN false
    clearError(), RETURN true
END FUNCTION

FUNCTION checkPasswordConfirm()
    IF password_confirm is empty THEN showError("Please re-enter your password."),
    RETURN false
```

IF password_confirm does not equal password_plain THEN showError("Passwords do not match."), RETURN false

clearError(), RETURN true

END FUNCTION

Privacy Acknowledgment

FUNCTION checkHipaa()

IF hipaa_ack is not checked THEN showError("Please acknowledge the privacy notice."), RETURN false

clearError(), RETURN true

END FUNCTION

4. When each check runs (on the fly)

All of these get hooked up in script.js when the page loads, so there is no JavaScript sitting inline in the HTML.

firstname, middleinit, lastname, preferred_language, addr1, addr2, city, email, member_id, userid: oninput clears the error as soon as the value becomes good, onblur runs the full check when the patient leaves the field.

dob: oninput runs formatDOB, onblur runs checkDOB.

ssn_plain: oninput runs formatSSN, onblur runs checkSSN.

zip and phone1: oninput strips anything that is not a number, onblur runs the full check.

state and insurance_provider: onchange.

sex: onchange runs checkRequiredRadio("sex").

has_insurance: onchange runs toggleInsurance and checkInsuranceProvider.

contact_pref: onchange re-runs checkPhone, since picking Phone or Text makes the phone number required.

hx_ checkboxes: onchange runs checkHealthHistory.

symptoms: oninput runs checkSymptoms and shows a live "characters left" count.

health_scale: oninput runs updateHealthScale.

password_plain: oninput runs checkPassword so the list of what is missing shrinks as they type, and re-runs checkPasswordConfirm if the confirm box already has something in it.

password_confirm: oninput runs checkPasswordConfirm.

userid: oninput also re-runs checkPassword, since the password cannot equal the User ID.

hipaa_ack: onchange runs checkHipaa.

Every field also has a title attribute and placeholder or hint text so the patient knows the rule before they start typing.

5. Buttons

VALIDATE replaces the old Submit button and runs validateForm.

The real Submit button starts out hidden.

IF the patient changes any field after a successful VALIDATE THEN

    HIDE the Submit button again (they have to re-validate so nothing bad slips through)

END IF

WHEN Submit is clicked THEN load thankyou.html, which says "Thank you for your submission. We will be contacting you shortly."

WHEN Clear Form is clicked THEN reset every field, clearError on every field, set health_scale_output back to 5, and HIDE the Submit button.